

SIMATS ENGINEERING ASSESSMENT TOOL 3

COURSE NAME: Artificial Intelligence

COURSE CODE: CSA17

COURSE OUTCOME COVERED

CO2: Experiment search and game-solving techniques such as Greedy Search, A, CSP, Minimax, and Alpha-Beta pruning with heuristic functions for solving complex AI problems efficiently. (BL3)*

Assessment Tool Weightage: 40%

QUESTIONS: 3

Duration: 1 Hour

Total Marks:30

Annexure A

Dry Run Evaluation

Code of Conduct:

(192511402)

I Bhuvaneshwaran J (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action

Signature of the student:



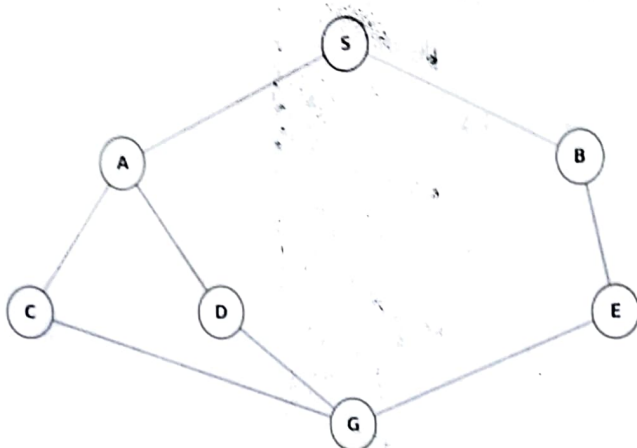
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Student 4: Bhuvaneshwaran J | Register Number: 192511402

1. Question 10 - A* versus Greedy Best-First Search

Use the displayed graph to perform Greedy Best-First Search and A* separately. Compare their expansion order and final path cost



Algorithm	Expansion Order	Path	Final Cost
1 Greedy Best First	$S \rightarrow A \rightarrow C \rightarrow G$	$h(n)$	3
2 A*	$g(n), h(n)$	$g(n), h(n)$	3

2. Question 11 - Hill Climbing

Consider the optimization values. 5 has neighbours 7 and 6, 7 has neighbours 5 and 10, 6 has neighbours 5 and 8, 8 has neighbours 6, 9 and 4.

Starting from 5, perform Hill Climbing to maximize the value.

3. Question 12 - Local Search / Optimization

A delivery company has routes $R_1=120$ km, $R_2=95$ km, $R_3=105$ km and $R_4=85$ km. The current solution is R_1 .

Perform a local-search improvement and explain local optimum versus global optimum.

Criteria	Weightage (%)	Level 4 - Exemplary (4)	Level 3 - Proficient (3)	Level 2 - Developing (2)	Level 1 - Beginning (1)
Understanding of Search Problem	20	Clearly understands the search problem, initial state, goal state, and search strategy	Understands the problem with minor misunderstandings	Demonstrates partial understanding of the search problem	Shows little or no understanding of the search problem
State Expansion and Dry Run Execution	30	Correctly traces every iteration, expands nodes accurately, and maintains the Queue/Stack/Priority Queue without errors	Performs most iterations correctly with minor mistakes in state expansion or data structure updates	Performs only some iterations correctly; several errors in tracing or node expansion	Unable to correctly perform the dry run or expand states
Application of Search Algorithm	20	Applies the selected algorithm (BFS/DFS/UCS) correctly, following all algorithmic rules	Applies the algorithm correctly with a few procedural errors	Applies only basic algorithm steps with noticeable mistakes	Fails to apply the correct search algorithm
Accuracy of Solution Path	20	Identifies the correct traversal order/solution path and goal state with proper justification	Solution path is mostly correct with minor errors	Partially correct solution path, goal may not be reached correctly	Incorrect or incomplete solution path
Presentation and Logical Reasoning	10	Queue/Stack/Priority Queue updates, state diagrams, and explanations are neat, organized, and logically presented	Presentation is clear with minor organizational issues	Limited explanation and organization, reasoning is partially clear	Poor presentation with unclear or missing reasoning

20 + 26 + 20 + 20 + 8

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fp

A* versus Greedy Best First Search

Start = S

Goal = G

Greedy Search:

$$f(n) = h(n)$$

It selects the node that appears closest to the goal according to the heuristic

Expansion :

S $\rightarrow$ A $\rightarrow$ C $\rightarrow$ G

Path :

S $\rightarrow$ A $\rightarrow$ C $\rightarrow$ G



Cost : 3

Each edges has cost 1, 3 edges.

(ii) A* Search

$$f(n) = g(n) + h(n)$$

$g(n)$ = cost from S to current node

$h(n)$ = estimated cost from current node to G

Assuming all edges:

$$S \rightarrow A \rightarrow D \rightarrow G$$

Final cost

$$1 + 1 + 1 = 3$$

Conclusion:

- * Greedy search considers only $h(n)$ while A* considers $g(n) + h(n)$
- * A* search gives a more reliable optimal solution when heuristic is admissible

Hill climbing:

Step 1: Start at 5

Current value :

5

Neighbours:

7, 6

choose the larger value

5 $\rightarrow$ 7

Step 2: At 7

Neighbours

5, 10

7 $\rightarrow$ 10

Step 3: At 10

no neighbours for 10

Path

5 $\rightarrow$ 7 $\rightarrow$ 10

Hill climbing always moves to a better neighbouring state. Starting from 5, it moves to 7 and then to 10, where no further move is given

Local Search

$$P_1 = 120 \text{ km}$$

$$P_2 = 45 \text{ km}$$

$$P_3 = 100 \text{ km}$$

$$P_4 = 85 \text{ km}$$

Step 3

$$P_1 = 120$$

Compare with the available alternatives

The best improvement is

$$P_4 = 85 \text{ km}$$

$$P_1 \rightarrow P_4$$

Improvement

$$120 - 85 = 35 \text{ km}$$

Final Solution

$$P_4 = 85 \text{ km}$$

Starting from P_1 , local search compared the available alternatives and choosed P_4 because it has the minimum distance. The improvement is 35 km. P_4 is the global optimum. Since there is no more improvement, the algorithm terminates.

Local Search:

$$R_1 = 120 \text{ km}$$

$$R_2 = 95 \text{ km}$$

$$R_3 = 100 \text{ km}$$

$$R_4 = 85 \text{ km}$$

Step 3

$$R_1 = 120$$

Compare with the available alternatives

The best improvement is

$$R_4 = 85 \text{ km}$$

$$R_1 \rightarrow R_4$$

Improvement:

$$120 - 85 = 35 \text{ km}$$

Final Solution

$$R_4 = 85 \text{ km}$$

Starting from R_1 , local search compares the available alternatives and chooses R_4 because it has the minimum distance. The improvement is 35 km. R_4 is the global optimum since the search space is small. Hence, the local search is sufficient.